Section 1.8 Trigonometric algebra & equations

三角恒等式与方程

[AIME-2002-II-Q10] | Part B (Q6-Q10)

Answer: 900

1. Equal sines mean the angles differ by $2j\pi$ or sum to $(2j+1)\pi$. Therefore the positive solution families are $360j\pi/(180-\pi)$, $j \geq 1$, and $180(2j+1)\pi/(180+\pi)$, $j \geq 0$.

2. The second family begins near 3.09 and the first near 6.39; the next second-family value is near 9.26. Thus the two least are the first members of these families.

3. The parameters are $360, 180, 180, 180$, summing to 900.

Source / 来源

Section 1.8 Trigonometric algebra & equations

三角恒等式与方程

[AIME-2008-I-Q08] | Part B (Q6-Q10)

Answer: 047

1. Tangent addition combines the first two angles to tangent $7/11$, then the first three to tangent $23/24$. The sums are acute, so no branch adjustment is needed.

2. The last angle has tangent $(1-23/24)/(1+23/24)=1/47$. Therefore $n=47$.

Source / 来源

Section 1.8 Trigonometric algebra & equations

三角恒等式与方程

[AIME-2008-II-Q08] | Part B (Q6-Q10)

Answer: 251

1. Product-to-sum writes each term as $\sin(k(k+1)a) - \sin((k-1)ka)$. The sum telescopes to $\sin(n(n+1)\pi/2008)$.

2. Integer sine values require an argument multiple of $\pi/2$, so $1004 = 4 \cdot 251$ divides $n(n+1)$. Coprimality of consecutive integers forces $n \geq 250$.

3. $250 \cdot 251$ lacks a factor four, while $251 \cdot 252$ qualifies. The minimum is 251.

Source / 来源

Section 1.8 Trigonometric algebra & equations

三角恒等式与方程

[AIME-2012-II-Q09] | Part B (Q6-Q10)

Answer: 107

1. The sine double-angle ratio is $3/2$. Let $t = \cos^2 y$; the two given squared relations sum to $9(1-t) + t/4 = 1$, so $t = 32/35$.

2. Thus $\cos 2y = 29/35$ and $\cos 2x = -19/35$. In particular neither required double-angle denominator vanishes.

3. The sum is $3/2 - 19/29 = 49/58$, giving 107.

Source / 来源

Section 1.8 Trigonometric algebra & equations

三角恒等式与方程

[AIME-2021-I-Q07] | Part B (Q6-Q10)

Answer: 063

1. Both sines must equal 1. Thus $mx = (4a+1)\pi/2$ and $nx = (4b+1)\pi/2$. Eliminating x gives $4(na - mb) = m - n$.

2. Let $g = \gcd(m, n)$, $m = gu, n = gv$. Integer a, b exist exactly when $u \equiv v \pmod 4$; coprimality then forces both odd.

3. For $g = 1, 2, 3, 4, 5, 6$, coprime pairs $u < v \le \lfloor 30/g \rfloor$ in the same odd residue class mod 4 number $44, 11, 4, 2, 1, 1$. For $g \ge 7$ there are none. The total is 63.

Source / 来源

Section 1.8 Trigonometric algebra & equations

三角恒等式与方程

[AIME-2025-II-Q09] | Part B (Q6-Q10)

Answer: 149

1. Zeros require $\sin 5x = k/7$, where $-7 \le k \le 7$ is integral. Over five full periods, $k = 0$ has nine zeros after excluding endpoints; each of twelve nonzero interior levels has ten; the two extrema have five each.

2. Thus $n = 9 + 120 + 10 = 139$. At a zero, $f\prime(x) = 35\pi \cos(k\pi) \cos 5x$ vanishes only at the ten extrema. These are genuine tangencies because the inner sine turns there. Hence $t = 10, n + t = 149$.

Source / 来源

Section 1.8 Trigonometric algebra & equations

三角恒等式与方程

[AIME-2000-II-Q15] | Part C (Q11-Q15)

Answer: 001

1. The identity $\cot k^\circ - \cot(k+1)^\circ = \sin 1^\circ / [\sin k^\circ \sin(k+1)^\circ]$ converts the sum, after multiplication by $\sin 1^\circ$, into an alternating cotangent sum from 45° through 134°.

2. Angles summing to 180° have equal parity and opposite cotangents, so their contributions cancel. Only $\cot 45^\circ = 1$ remains; the central term $\cot 90^\circ$ is zero.

3. Therefore the original sum is $1/\sin 1^\circ$. The positive integer $n = 1$ works and is already the smallest possible.

Source / 来源

Section 1.8 Trigonometric algebra & equations

三角恒等式与方程

[AIME-2006-I-Q12] | Part C (Q11-Q15)

Answer: 906

1. Use $2\cos 4x\cos x=\cos 3x+\cos 5x$. Setting these latter cosines to u,v turns the equation into $u^3+v^3=(u+v)^3$, so $uv(u+v)=0$.

2. Thus zeros come from $\cos 3x=0$, $\cos 5x=0$, $\cos 4x=0$, or $\cos x=0$. In the given interval their distinct solutions are $150;126,162,198;112.5,157.5;$ and none, respectively.

3. The sum is 906.

Source / 来源

Section 1.8 Trigonometric algebra & equations

三角恒等式与方程

[AIME-2013-II-Q15] | Part C (Q11-Q15)

Answer: 222

1. For triangle angles, the sine and cosine laws give $\sin^2 A + \sin^2 B - 2\sin A \sin B \cos C = \sin^2 C$. Thus each displayed expression equals two minus the squared sine of the omitted angle.

2. Therefore $\sin^2 C = 1/8$ and $\sin^2 A = 4/9$. Acuteness fixes $\sin A = 2/3, \cos A = \sqrt{5}/3, \sin C = \sqrt{2}/4, \cos C = \sqrt{14}/4$.

3. Since $B = \pi - A - C$, $\sin B = (2\sqrt{14} + \sqrt{10})/12$, with square $(33 + 4\sqrt{35})/72$. The resulting $A + C$ is acute, consistent with obtuse B.

4. The target is $2 - \sin^2 B = (111 - 4\sqrt{35})/72$. As 115 and 72 are coprime, the requested sum is 222.

Source / 来源

Section 1.8 Trigonometric algebra & equations

三角恒等式与方程

[AIME-2014-II-Q12] | Part C (Q11-Q15)

Answer: 399

1. Using $A+B+C=\pi$ and sum-to-product identities gives $1-(\cos 3A+\cos 3B+\cos 3C)=4\sin(3A/2)\sin(3B/2)\sin(3C/2)$.

2. The product is zero. For a triangle angle $0<X<\pi$, the only possible zero of $\sin(3X/2)$ is $X=120^\circ$, so one angle must be one hundred twenty degrees. Conversely such an angle satisfies the identity.

3. The largest possible third side puts this angle between the known sides, giving squared length $10^2+13^2+130=399$. If the obtuse angle is opposite a known side, the third side is shorter than that side and cannot exceed this value.

4. Thus $m=399$.

Source / 来源

Section 1.8 Trigonometric algebra & equations

三角恒等式与方程

[AIME-2015-I-Q13] | Part C (Q11-Q15)

Answer: 091

1. Let P be the product of sines at odd angles $1^\circ, 3^\circ, \ldots, 89^\circ$, and Q at even angles $2^\circ, 4^\circ, \ldots, 88^\circ$. The product of all sines from 1° to 89° is PQ; replacing every angle by its complement gives the same product of cosines.

2. Thus $(PQ)^2 = \prod_{j=1}^{89} \sin j^\circ \cos j^\circ = 2^{-89} \prod_{j=1}^{89} \sin(2j)^\circ = 2^{-89} Q^2$, because reflection about 90° pairs all even-angle factors and the 90° factor is 1.

3. Cancel the nonzero Q^2 to get $P^{-2} = 2^{89}$. As 89 is prime, the only integers $m, n > 1$ giving this power are $m = 2, n = 89$. Their sum is 91.

Source / 来源

Section 1.8 Trigonometric algebra & equations

三角恒等式与方程

[AIME-2015-II-Q13] | Part C (Q11-Q15)

Answer: 628

1. The identity $2\sin(1/2)\sin k = \cos(k-1/2) - \cos(k+1/2)$ telescopes to $a_n = [\cos(1/2) - \cos(n+1/2)]/[2\sin(1/2)]$.

2. The denominator is positive. Negativity is equivalent to $n+1/2$ lying within $1/2$ of a multiple $2\pi m$, that is $2\pi m - 1 < n < 2\pi m$.

3. For each positive integer m, this interval contains exactly one integer, $\lfloor 2\pi m \rfloor$, because $2\pi m$ is irrational. Their order increases with m, so the 100th is $\lfloor 200\pi \rfloor = 628$.

Source / 来源

Section 1.8 Trigonometric algebra & equations

三角恒等式与方程

[AIME-2022-I-Q15] | Part C (Q11-Q15)

Answer: 033

1. Real radicands imply $0 < x, y, z \leq 2$. Set $x = 2\sin^2 a, y = 2\sin^2 b, z = 2\sin^2 c$, where $a, b, c \in (0^\circ, 90^\circ]$. The equations become $\sin(a+b) = 1/2$, $\sin(b+c) = \sqrt{2}/2$, and $\sin(c+a) = \sqrt{3}/2$.

2. The possible pair-sums are $a+b \in \{30, 150\}$, $b+c \in \{45, 135\}$, $c+a \in \{60, 120\}$ degrees. Of their eight combinations, only $(30, 45, 60)$ and $(150, 135, 120)$ yield all three angles in the stated range. They give $(a, b, c) = (22.5, 7.5, 37.5)$ or $(67.5, 82.5, 52.5)$.

3. In the first case, the product is $\cos 45^\circ \cos 15^\circ \cos 75^\circ = \sqrt{2}/8$; in the second each cosine changes sign, so the product is its negative. The square is always $1/32$, giving 33.

Source / 来源

Section 1.8 Trigonometric algebra & equations

三角恒等式与方程

[AIME-2023-II-Q13] | Part C (Q11-Q15)

Answer: 167

1. Set $s = \sec A, t = \tan A$. Then $st = 2, s^2 - t^2 = 1$, so for $u = s^2, v = t^2$, one has $u + v = \sqrt{17}, uv = 4$.

2. The sums $w_j = u^j + v^j$ satisfy $w_{j+1} = \sqrt{17}w_j - 4w_{j-1}$, with $w_0 = 2, w_1 = \sqrt{17}$. Induction shows that even-index terms are positive integers and odd-index terms are nonzero integer multiples of $\sqrt{17}$. For odd n, $(s^n + t^n)^2 = w_n + 2^{n+1}$ is irrational. Thus integrality occurs exactly when $4 \mid n$.

3. For $n = 4j$, let $x_j = s^{4j} + t^{4j}$. The two bases have sum 9 and product 16, so $x_{j+1} = 9x_j - 16x_{j-1}$ with $x_0 = 2, x_1 = 9$.

4. Modulo 10, the terms for $j \geq 1$ cycle $9, 9, 7$: the pair $(x_1, x_2) = (9, 9)$ returns after three steps. Valid $j \leq 250$ are exactly those not divisible by 3, numbering $250 - 83 = 167$.

Source / 来源